

# Skills Summary

## Out of What? And Who Got Asked?

Use this reference while completing your worksheet or when studying.

### Skill 1 — Choosing the right denominator

**Rule:**  $P(\text{event}) = \frac{\text{number of outcomes you want}}{\text{number of outcomes altogether}}$

- The denominator counts *everything that could happen*, including the outcomes you want. Never the leftovers on their own.
- If something is removed and not replaced, the total changes too — so re-check both numbers, not just the one the story mentions.
- Check: all the separate probabilities for one draw must add to 1.

**Example:** A drawer holds 7 red socks and 13 blue socks. One is taken without looking. Find  $P(\text{red})$ .

**Step 1.** Settle “out of what?” first: one sock is taken from the whole drawer, so the denominator counts every sock, red ones included.

**Step 2.** Altogether there are  $7 + 13 = 20$  socks, and 7 of them are red, so  $P(\text{red}) = \frac{7}{20}$ .

**Try it:** A jar holds 9 mints and 6 toffees. One sweet is taken without looking. Find  $P(\text{toffee})$  in lowest terms.

$\frac{9}{20}$  :æp

**(!) Watch out:**  $\frac{7}{13}$  compares red socks with blue socks. That is a *ratio*, not a probability, and it is always too big.

### Skill 2 — Deciding whether a sample is fair

**The test:** could *every* member of the group you want to describe have been picked?

- If where or how you asked rules some people out, the sample is **biased** — and a bigger biased sample is still biased.
- Say which way it is wrong: asking at the bike rack over-states cycling.
- Scaling up a *fair* sample:  $\text{estimate} = \text{population} \times \frac{\text{count in sample}}{\text{size of sample}}$ .

**Example:** 25 students are drawn at random from a school roll of 400, and 10 of them walk to school. Estimate how many of the 400 walk.

**Step 1.** Names were drawn at random from the roll, so every student could have been picked — the sample passes the fairness test, and only then is it worth scaling up.

**Step 2.** The sample fraction is  $\frac{10}{25}$ , so the estimate is  $400 \times \frac{10}{25} = \mathbf{160}$  students.

**Try it:** 40 students are drawn at random from a roll of 600, and 12 of them play an instrument. Estimate how many of the 600 do.

check: 081

### Skill 3 — Finding a median from a sample

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**Rule:** sort the values smallest to largest *first*, then take the middle one.

- An odd number of values has one middle value.
- An even number of values has two: the median is halfway between them.
- The median describes only the group you sampled. A median from a biased sample is a correct number about the wrong group.

**Example:** Find the median of 9, 4, 12, 7, 5, 10.

**Step 1.** A median is a position in the sorted order, so nothing can be read off the list as written — sort it before looking for the middle.

**Step 2.** Sorted: 4, 5, 7, 9, 10, 12. There are six values, so take the mean of the third and fourth:  $\frac{7+9}{2} = 8$ .

**Try it:** Find the median of 6, 2, 9, 3, 8, 4, 7.

check: 9